

- Functional Requirements Document (FRD) for SCDType2Application**
- Introduction:**

The SCDType2Application is designed to integrate and aggregate customer and order data from CSV files, implementing SCD type 2 logic and generating aggregated reports.

Functional Requirements

1. **Data Ingestion Functionality:**

- The system shall read source CSV data from specified volume locations (`/Volumes/gen_ai_poc_databrickscoe/sdlc_wizard/customerdata`` and `/Volumes/gen_ai_poc_databrickscoe/sdlc_wizard/orderdata``) into Delta tables for customer and order data.
- The system shall create Delta tables with the specified schema:

+ Customer table: ``CustId`, `Name`, `EmailId`, `Region``.

+ Order table: ``OrderId`, `ItemName`, `PricePerUnit`, `Qty`, `Date`, `CustId``.

2. **Data Cleansing Functionality:**

- The system shall remove "Null"/null and duplicate records from both the customer and order tables.

3. **Data Transformation Functionality:**

- The system shall create an "ordersummary" table if it does not exist in the catalog "gen_ai_poc_databrickscoe" and schema "sdlc_wizard" with the following schema: ``CustId`, `Name`, `EmailId`, `Region`, `OrderId`, `ItemName`, `PricePerUnit`, `Qty`, `Date`, `StartDate`, `EndDate`, `IsActive``.
- The system shall join customer and order data using the "CustId" field and load the data into the "ordersummary" SCD type 2 table.

4. **SCD Type 2 Implementation:**

- The system shall implement SCD type 2 logic to update the "ordersummary" table whenever there is a change in the customer table.
- The system shall make old records inactive and new records active.
- The system shall update ``StartDate`` and ``EndDate`` accordingly.

5. **Aggregated Reporting Functionality:**

- The system shall create a new table "customeraggregatespend" with columns "Name", "TotalAmount", and "Date" in catalog "gen_ai_poc_databrickscoe" and schema "sdlc_wizard" if it does not exist.
- The system shall aggregate the total amount spent by customers from the "ordersummary" table and group by "Name" and "Date" columns.
- The system shall load the aggregated data into the "customeraggregatespend" table.

Data Flow Diagram (DFD)

```
```python
```

```
dfd_text = """
```

1. Read customer and order data from CSV files.
2. Load data into Delta tables (customer and order tables).
3. Remove null/null and duplicate records from customer and order tables.
4. Join customer and order data using CustId and load into ordersummary SCD type 2 table.
5. Implement SCD type 2 logic to update ordersummary table on customer data changes.
6. Aggregate total amount spent by customers from ordersummary table.
7. Load aggregated data into customeraggregatespend table.

```
"""
```

```
print(generate_dfd_diagram(dfd_text))
```

```
```
```

System Design

The system will be built using Databricks and Delta tables. The data ingestion, cleansing, transformation, and aggregation will be implemented using Spark SQL and Python.

Interface Requirements

- Input: CSV files containing customer and order data.
- Output: Delta tables (customer, order, ordersummary, and customeraggregatespend).

Data Requirements

- The system will store data in Delta tables.
- The schema for the Delta tables is specified in the functional requirements.

Acceptance Criteria

1. The data is successfully ingested into Delta tables.
2. The data is cleansed and transformed as required.
3. The SCD type 2 logic is implemented correctly.
4. The aggregated data is accurately calculated and loaded into the "customeraggregatespend" table.

By following this FRD, the development team should be able to design and implement the SCDType2Application that meets the business requirements.